

Introduction to Machine Learning

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Outline

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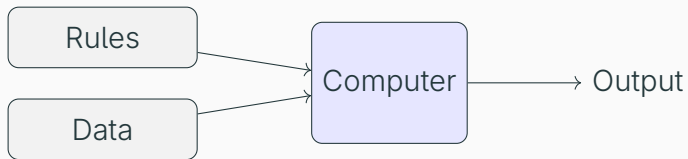
Introduction to Machine Learning

What is Machine Learning?

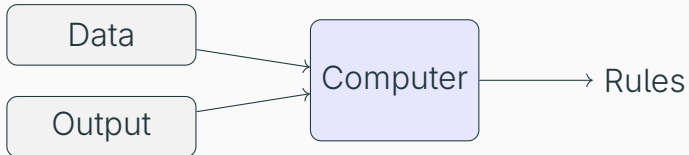
Tom Mitchell's formal definition: a program learns from experience E with respect to task T and performance measure P , if its performance at T , measured by P , improves with E .

- Instead of a human writing explicit rules, we give the algorithm examples (data), and it discovers the rules (a **model**) on its own.
- Example: a spam filter written the traditional way needs rules like “block if subject contains ‘free money’.” An ML spam filter is instead shown thousands of emails labeled spam / not spam, and it learns the patterns itself — including patterns a human would never think to hard-code.
- This is the key shift: traditional programming is *rules + data* $\rightarrow$ *output*; machine learning is *data + output* $\rightarrow$ *rules*.

Traditional programming

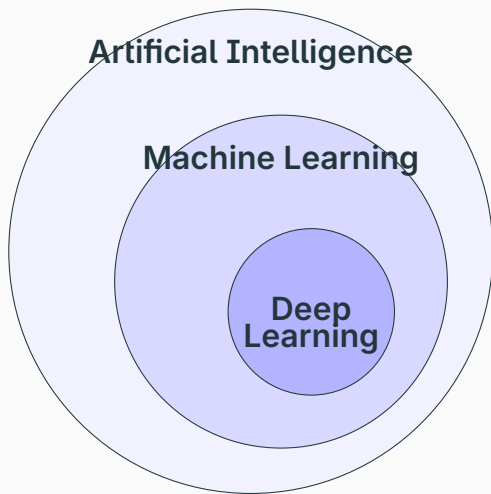


Machine learning



Scope: AI, Machine Learning, and Deep Learning

- **Artificial Intelligence (AI)**: any technique that lets a machine mimic intelligent behavior — this includes hand-coded expert systems (e.g., a chess engine with programmed rules) and does not require learning from data at all.
- **Machine Learning (ML)**: the subset of AI where the system *learns* its behavior from data rather than being explicitly programmed.
- **Deep Learning (DL)**: the subset of ML that uses multi-layered neural networks, especially effective for unstructured data like images, audio, and text (e.g., ChatGPT, face recognition on your phone).



Expert systems,
search & planning,
robotics rules

↪ only in AI, not ML

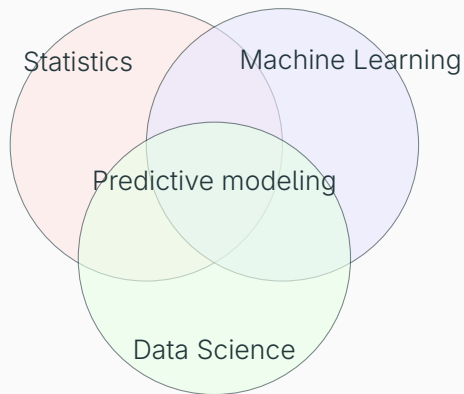
Machine Learning, Statistics, and Data Science

- **Statistics:** the mathematical discipline of collecting, analyzing, and interpreting data, with heavy emphasis on *inference* — quantifying uncertainty, testing hypotheses, estimating confidence intervals.
- **Data Science:** a broad, applied field that combines domain knowledge, programming/engineering, statistics, and ML to go from raw data to business decisions — includes data collection, cleaning, visualization, and communicating results to stakeholders.
- **Machine Learning:** algorithms that learn patterns from data primarily to make *predictions* on new, unseen data.

Machine Learning, Statistics, and Data Science (cont.)

- Statistics supplies much of ML's theoretical foundation (linear regression, maximum likelihood estimation, hypothesis testing).
- ML is one tool a data scientist uses, alongside dashboards, ETL pipelines, and A/B testing.
- A statistician cares whether a coefficient is significant and interpretable; an ML engineer mostly cares whether the model predicts well on new data, even if it is a black box.
- A data scientist at a company typically needs a bit of both statistics and ML, as well as the ability to explain results to a non-technical manager.

Machine Learning, Statistics, and Data Science (cont.)



Types of Machine Learning

Supervised Learning

Learning a mapping from input **features** X (the measurable inputs, e.g., square footage, pixel values) to an output **label** Y (the “correct answer”), using a **training set** of examples where Y is already known.

- **Classification:** label is a discrete category. Example: is this email spam or not? Does this X-ray show a tumor or not (used in hospital diagnostic - support tools)?
- **Regression:** label is a continuous number. Example: predicting a house's sale price from its size, location, and age; predicting tomorrow's temperature.
- The **model** is the learned function $\hat{Y} = f(X)$; a **loss function** measures how wrong its predictions are, and training adjusts the model to minimize this loss on the training data while still **generalizing** well to new, unseen data (avoiding **overfitting**, i.e., memorizing training data instead of learning general patterns).

Supervised Learning: Examples

- Everyday examples: your email's spam filter, your phone unlocking with face recognition, autocorrect/predictive text.
- Industry examples: banks predicting loan default risk (credit scoring), Tesla/Waymo detecting pedestrians and stop signs from camera images, hospitals using ML to flag likely - cancerous scans for radiologist review.

Unsupervised Learning

Finding structure or patterns in data that has **no labels** — there is no “correct answer” provided, the algorithm must discover structure on its own.

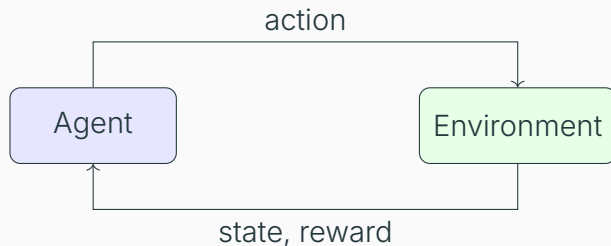
- **Clustering**: grouping similar data points together (e.g., k -means). Example: a retailer segments customers into groups by purchasing behavior, without being told in advance what the groups should be, in order to target each group with different marketing.
- **Dimensionality reduction**: compressing many features into a few that capture most of the information (e.g., PCA), useful for visualization or as a preprocessing step.
- **Anomaly detection**: flagging data points that don't fit the usual pattern. Example: banks flagging an unusual credit card transaction as possible fraud.

Unsupervised Learning: Examples

- Everyday example: Google Photos automatically grouping photos of the same person's face together, even though it was never told who that person is.
- Industry example: genomics researchers clustering gene-expression data to discover previously unknown subtypes of a disease.

Reinforcement Learning

An **agent** learns to make a sequence of decisions by interacting with an **environment**: at each **state**, it takes an **action**, receives a **reward** (or penalty), and updates its **policy** (its strategy for choosing actions) to maximize cumulative reward over time.



Reinforcement Learning (cont.)

- Unlike supervised learning, there is no fixed “correct” action for each state — the agent must learn from trial and error, and rewards can be delayed (an action now may only pay off many steps later).
- Analogy: training a dog with treats — you don’t tell it the exact muscle movements for “sit,” you reward it when it happens to get closer to the desired behavior.
- Example: DeepMind’s AlphaGo learned to play Go well enough to beat the world champion, largely through self-play reinforcement learning.

- Robots learning to walk or grasp objects through repeated physical trial and error.
- Google used reinforcement learning to reduce the energy used for cooling its data centers.
- YouTube uses RL -based methods to optimize recommendations for long-term viewer engagement rather than just the next click.

Comparing the Three Paradigms

	Supervised	Unsupervised	Reinforce- ment
Data	Labeled exam- ples	Unlabeled data	Interaction with environ- ment
Goal	Predict a known target	Discover hid- den structure	Maximize cumulative reward
Example	Spam detection	Customer seg- mentation	Game- playing agent, robotics

Applications of Machine Learning

Applications in Everyday Life

- **Voice assistants:** Siri, Alexa, and Google Assistant use ML for speech recognition and natural language understanding.
- **Recommendations:** Netflix suggesting what to watch, Spotify's Discover Weekly, YouTube's homepage, Amazon's "customers also bought."
- **Navigation:** Google Maps predicting traffic and estimated arrival time from historical and real-time location data.
- **Photography and communication:** face unlock on smartphones, autocorrect and predictive text, spam filtering in email, automatic photo tagging.
- **Social media:** automatically filtering out content, ranking your feed, and suggesting people you may know.

Applications in Industry and Organizations

- **Healthcare:** diagnostic support from medical images (X-rays, MRIs); DeepMind's AlphaFold predicting 3D protein structures, accelerating drug discovery.
- **Finance:** fraud detection on card transactions, algorithmic trading, automated credit scoring for loan approval.
- **Manufacturing:** predictive maintenance — companies like Siemens and GE use sensor data to predict when a machine will fail *before* it breaks down, avoiding costly downtime.
- **Retail:** demand forecasting and inventory management (e.g., Amazon predicting how much of a product to stock in each warehouse).

Applications in Industry and Organizations (cont.)

- **Transportation:** self-driving vehicles (Tesla, Waymo), and dynamic/surge pricing for ride-sharing (Uber, Lyft) based on real-time demand.
- **Agriculture:** predicting crop yields and detecting plant disease from drone/satellite imagery for precision farming.

Questions?
